

Run Report: `make all` Output and Discussion

TEEP Internship

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Abstract

This document walks through the literal console output of one full `make all` run (validate → split → eval-harness → test → stats → plots → baselines, run 2026-08-19, all 10 stages completed with no errors) and discusses what each stage’s numbers mean. Where `report/report.tex` presents the polished, aggregated narrative, this document keeps the per-fold detail that report intentionally compresses away, since that detail turns out to carry a real finding: fold composition, not separation method, explains most of the fold-to-fold variance in the results. As in `report.tex`, every number here is computed on the current, provisional (GitHub-sourced, not yet Mendeley-verified) copy of HLS-CMDS.

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1 Stage 1: Dataset Validation (`make validate`)

`load_dataset.py` loaded all three manifests and resolved every referenced audio file:

```

HS: 50 rows, all audio files found
LS: 50 rows, all audio files found
Mix: 145 rows, all audio files found
Sample rate: 4000 Hz
Additivity check (mixed ~= a*(heart+lung)): 36/145 rows additive

```

Discussion. The first three lines are a pure integrity check — every row’s audio exists and is readable — and are unremarkable on their own. The last two lines are not: 4000 Hz does not match the 22,050 Hz the HLS-CMDS descriptor paper documents, and only 36 of 145 mixture rows are acoustically related to their own named heart/lung sources. Both are symptoms of the same root cause (this working copy is sourced from a GitHub mirror, not the Mendeley release the project specifies) and both propagate into every later stage’s numbers. This is the single most important fact to carry forward from this run: nothing below should be read as a claim about HLS-CMDS itself until it is re-verified on the correct dataset copy.

2 Stage 2: Leakage-Safe Split (make split)

```

Leak groups: 29 covering 145 mix rows
Group sizes: [32, 22, 19, 13, 11, 9, 4, 4, 3, 3, 3, 2, 2, 2,
              1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]

```

Table 1: 5-fold split (seed 0) over the full 145-row `Mix.csv`.

Fold	Eval rows	HS pool	LS pool
0	32	44/50	46/50
1	29	43/50	46/50
2	28	44/50	47/50
3	28	40/50	43/50
4	28	47/50	38/50

Discussion. Fold 0 is visibly larger (32 rows) than the other four (28–29) because it absorbs the single size-32 leak group whole — splitting it would violate the no-leakage guarantee, so the greedy largest-group-first assignment accepts the imbalance instead. This is expected and by design, but it means fold 0’s per-fold numbers (Sections 3–7) are averaged over more rows than any other fold and should not be read as “a random 20% of the data” the way the other four folds approximately are.

3 Stage 3: Evaluation Harness Smoke Test (make eval-harness)

This stage runs a no-op “baseline” (returns the mixed signal unmodified as both source estimates) through the full k-fold harness purely to confirm the fold → fit → separate → score → aggregate plumbing works end-to-end against real audio, before any real separation method is trusted. Because it applies no processing, its numbers also double as the “no-separation” reference used in `report.tex`.

Table 2: No-separation reference, per fold (full 145 rows).

Fold	Heart			Lung		
	SDR	SIR	SAR	SDR	SIR	SAR
0	-14.93	1.78	0.99	-17.14	-1.61	0.99
1	-14.78	0.40	5.04	-15.56	-0.24	5.04
2	-12.05	2.79	10.30	-15.78	-2.58	10.30
3	-12.77	2.33	6.61	-16.13	-2.16	6.61
4	-13.89	1.24	9.36	-14.85	-1.17	9.36
Mean $\pm$ std	-13.68 ± 1.26	1.71 ± 0.93	6.46 ± 3.71	-15.89 ± 0.84	-1.55 ± 0.91	6.46 ± 3.71

Discussion. Two sanity properties confirm the harness is wired correctly: SAR is identical for heart and lung within every fold (with no processing applied, there is nothing to distinguish one source’s “artifacts” from the other’s — both estimates are the same signal), and SAR moves in lockstep with fold size/composition (fold 0’s 0.99 dB SAR is far below the others’ 5–10 dB), which is exactly the pattern Section 7 shows persisting once a real separation method is applied. That persistence is the tell that fold identity, not method choice, is doing a lot of the work in these numbers — a theme this report returns to in Section 8.

4 Stage 4: Test Suite (`make test`)

31 passed, 3 warnings in 114.04s (0:01:54)

Discussion. All 31 tests across `test_metrics.py` (BSS Eval wrapper correctness), `test_split.py` (leakage-safety of the fold assignment), and `test_load_dataset.py` (closed-form and real-data checks on the additive-triplet audit) pass. The 3 warnings are third-party `DeprecationWarnings` from `audioread`’s use of three deprecated Python standard-library modules (`aifc`, `audioop`, `sunau`), unrelated to this project’s code. This is the evidence that the numbers in Sections 3 and 7 are at least *computed correctly* (metrics match a direct `mir_eval` call, no held-out recording leaks into any fold’s dictionary pool) — it says nothing about whether the dataset those numbers are computed *on* is correct (Section 1’s caveat still applies).

5 Stage 5: Dataset Statistics (`make stats`)

Scanned 535 audio files

Sample rate(s): [4000] Hz Channel count(s): [1]
Duration: mean=15.00s min=15.00s max=15.00s
Clipped files: 36 (6.7%)

Per-class and per-location breakdown (isolated `HS.csv`/`LS.csv` recordings only; duration and sample rate are constant everywhere, so omitted below):

Table 3: Per-class counts, all with 0 clipped files.

Heart class	n	Lung class	n
Normal	9	Normal	12
Mid Systolic Murmur	7	Coarse Crackles	9
Late Diastolic Murmur	6	Pleural Rub	9
Early Systolic Murmur	6	Rhonchi	8
Late Systolic Murmur	5	Wheezing	7
S3	5	Fine Crackles	5
Atrial Fibrillation	4		
AV Block	3		
Tachycardia	3		
S4	2		

Discussion. This output is discussed at length in `report.tex` §2.2.1 (why it is trustworthy, and the class-imbalance caveat it surfaces for the planned classification stage — S4 at $n=2$ cannot appear in every fold’s training set). The one point worth repeating here: every class and every location shows *zero* clipped files, which is the finer-grained confirmation that the 36 clipped files counted above are entirely confined to `Mix.csv`’s mixed component — the same 36 rows the additive-triplet audit (Section 1) independently flags as valid, i.e. the clipping is a normalization fingerprint of a real heart+lung summation, not a defect scattered across the dataset.

6 Stage 6: Visualization (make plots)

Wrote 4 figures to `src/visualization/plots/`

Discussion. The four visualization scripts — `audio_plotter`, `audio_spectrogram`, `donut_chart`, `plot_per_class` — all ran without error under `MPLBACKEND=Agg` (headless). This stage produces gitignored PNGs consumed visually, not numbers to discuss — included here only to record that it completed as part of this run.

7 Stage 7: Separation Baselines (make baselines)

The slow stage (per-fold NMF dictionary fitting), and the one that produces this run’s headline result. Both baselines are run on both the full 145-row set and the 36-row additive-valid subset. **The two row sets use independently computed fold assignments** (`eval_harness.py`’s `mix_df` parameter reassigns folds within whatever subset it is given), so “fold 0” under the full set and “fold 0” under the valid subset are *not* the same rows — only Baseline 1 and Baseline 2 *within the same row set* share fold identity (same seed, same `mix_df`), and only those pairs are safe to compare fold-by-fold below.

7.1 Baseline 1 (bandpass): full 145 rows

Table 4: Baseline 1, full 145 rows, per fold.

Fold	Heart			Lung		
	SDR	SIR	SAR	SDR	SIR	SAR
0	-13.85	4.30	-9.86	-15.25	1.73	-11.40
1	-13.93	2.36	-9.12	-14.15	1.85	-9.51
2	-10.74	5.44	-6.23	-12.45	2.15	-8.18
3	-11.69	4.61	-7.33	-14.33	1.49	-9.48
4	-13.04	3.61	-7.95	-12.96	1.43	-7.79
Mean $\pm$ std	-12.65 $\pm$ 1.40	4.07 $\pm$ 1.16	-8.10 $\pm$ 1.44	-13.83 $\pm$ 1.12	1.73 $\pm$ 0.29	-9.27 $\pm$ 1.42

Table 5: Baseline 1, full 145 rows, pooled row-level statistics.

Source	SDR		SIR		SAR	
	mean	median	mean	median	mean	median
Heart	-12.69	-18.39	4.06	3.61	-8.15	-16.52
Lung	-13.87	-19.19	1.73	1.70	-9.33	-16.01

7.2 Baseline 1 (bandpass): 36-row additive-valid subset

Table 6: Baseline 1, 36-row valid subset, per fold.

Fold	Heart			Lung		
	SDR	SIR	SAR	SDR	SIR	SAR
0	0.38	1.21	13.04	7.96	9.42	16.79
1	8.67	9.83	20.86	4.28	6.55	10.53
2	8.19	8.74	18.97	4.30	6.24	10.47
3	3.09	3.75	16.94	5.01	6.39	13.04
4	7.70	8.70	18.17	-1.83	-0.49	10.61
Mean $\pm$ std	5.61 $\pm$ 3.68	6.45 $\pm$ 3.76	17.60 $\pm$ 2.92	3.94 $\pm$ 3.57	5.62 $\pm$ 3.66	12.29 $\pm$ 2.74

Table 7: Baseline 1, 36-row valid subset, pooled row-level statistics.

Source	SDR		SIR		SAR	
	mean	median	mean	median	mean	median
Heart	5.46	5.18	6.30	5.78	17.47	19.13
Lung	4.05	5.05	5.73	5.85	12.42	12.85

7.3 Baseline 2 (supervised NMF): full 145 rows

Table 8: Baseline 2, full 145 rows, per fold.

Fold	Heart			Lung		
	SDR	SIR	SAR	SDR	SIR	SAR
0	-13.76	4.16	-10.93	-16.57	-0.02	-11.19
1	-14.29	1.78	-9.91	-15.41	0.84	-10.08
2	-10.92	6.13	-8.55	-14.68	-0.53	-7.66
3	-12.47	5.15	-9.33	-14.82	0.52	-8.98
4	-12.64	3.89	-9.18	-13.95	0.77	-8.27
Mean $\pm$ std	-12.82 $\pm$ 1.31	4.22 $\pm$ 1.63	-9.58 $\pm$ 0.90	-15.09 $\pm$ 0.98	0.32 $\pm$ 0.58	-9.23 $\pm$ 1.42

Table 9: Baseline 2, full 145 rows, pooled row-level statistics.

Source	SDR		SIR		SAR	
	mean	median	mean	median	mean	median
Heart	-12.85	-17.76	4.20	3.07	-9.62	-16.10
Lung	-15.13	-19.21	0.31	0.63	-9.29	-16.19

7.4 Baseline 2 (supervised NMF): 36-row additive-valid subset

Table 10: Baseline 2, 36-row valid subset, per fold.

Fold	Heart			Lung		
	SDR	SIR	SAR	SDR	SIR	SAR
0	-0.56	1.86	8.23	4.72	6.01	14.05
1	5.87	11.10	10.93	-1.75	-0.96	12.29
2	6.79	10.96	9.81	-1.58	-1.09	12.49
3	1.44	3.08	10.63	1.75	2.70	13.68
4	5.36	9.72	9.22	-5.33	-4.89	12.58
Mean $\pm$ std	3.78 $\pm$ 3.17	7.34 $\pm$ 4.50	9.77 $\pm$ 1.09	-0.44 $\pm$ 3.82	0.35 $\pm$ 4.15	13.02 $\pm$ 0.79

Table 11: Baseline 2, 36-row valid subset, pooled row-level statistics.

Source	SDR		SIR		SAR	
	mean	median	mean	median	mean	median
Heart	3.66	5.66	7.19	8.66	9.72	9.21
Lung	-0.29	1.34	0.51	1.84	13.05	13.16

7.5 Cross-baseline fold correlation

Because Baseline 1 and Baseline 2 use the same fold assignment within each row set (Section 7), their per-fold numbers are directly comparable fold-by-fold. Ranking the full-145 folds by heart SDR:

- Baseline 1: fold 2 (best) > fold 3 > fold 4 > fold 0 > fold 1 (worst)
- Baseline 2: fold 2 (best) > fold 3 > fold 4 > fold 0 > fold 1 (worst)

The order is *identical*. The same comparison on the 36-row valid subset:

- Baseline 1: fold 1 (best) > fold 2 > fold 4 > fold 3 > fold 0 (worst)
- Baseline 2: fold 2 (best) > fold 1 > fold 4 > fold 3 > fold 0 (worst)

Folds 1 and 2 swap at the top, but fold 0 is worst and fold 3 is second-worst in both. Lung SDR shows the same qualitative pattern with fold 0 worst in three of the four method/row-set combinations (the exception is Baseline 2 on the full set, where fold 0 is worst by an even larger margin).

Discussion. With only 5 folds this is not a statistically rigorous claim, but the consistency is striking: *which fold a row lands in* predicts separation quality about as well as *which of the two methods* processed it. The most direct explanation is that folds differ in how many genuinely additive rows they contain (Section 1) and/or in which heart/lung classes dominate them, and that this per-fold composition effect is larger than the gap between a zero-training filter and a fitted NMF dictionary. This is a caution for any future baseline comparison on this dataset: a single fold’s number is not informative about method quality on its own, and a new method should be checked for the same cross-fold ranking pattern before its aggregate score is trusted.

8 Discussion

Pulling the seven stages together:

1. **The pipeline itself is healthy.** All 10 make all targets completed without error, all 31 tests pass, and the harness’s plumbing smoke test (Section 3) confirms the fold $\rightarrow$ fit $\rightarrow$ score $\rightarrow$ aggregate path works correctly end-to-end. Nothing in this run indicates a code-level bug.
2. **The dataset is not yet trustworthy.** 109 of 145 mixture rows have no genuine heart+lung structure to separate (Section 1), and the sample rate itself is off by $5.5\times$ from the documented value — both symptoms of sourcing this working copy from GitHub instead of Mendeley. Every quantitative result below the dataset caveat is provisional.
3. **Restricting to genuinely valid rows changes the story, not just the scale.** SDR jumps 15–18 dB once the 109 invalid rows are excluded (Section 7) — expected, since those rows have nothing coherent to separate — but more importantly, the *ranking* of the two baselines flips from a near-tie (full set) to a clear, consistent win for the zero-training bandpass filter over learned NMF (valid subset). A comparison run only on the full, mostly-invalid set would have reported the wrong qualitative conclusion, not just a noisier version of the right one.
4. **Fold composition is a bigger effect than it looks.** Section 7.5’s fold-ranking agreement between two unrelated separation methods suggests a meaningful share of the fold-to-fold variance reported as “std” in `report.tex`’s aggregate tables reflects which rows are in each fold, not method instability. This does not change any reported number, but it does change how those numbers should be read: as method comparisons that are only as reliable as the row composition they share, not as independent per-fold measurements of method quality.

9 Next Steps

- Re-download HLS-CMDS from Mendeley and re-run this exact pipeline on the correct copy — the blocking action item carried over from `report.tex`'s Limitations section.
- Re-run Section 7.5's fold-ranking check on the Mendeley copy once the split changes, to see whether the composition-dominates-method pattern is a property of this specific (small, imbalanced) row set or persists more generally.
- Feed this run's per-class counts (Section 5) into the classification stage design (`PROTOCOL.md` §5.3) before it is built, given the S4/AV Block/Tachycardia class-size caveat already flagged in `report.tex`.